



**SCHOOL OF MATHEMATICS,
APPLIED STATISTICS & ANALYTICS**

A Parametric Approach to Quantifying Risk and Uncertainty in Indian Market-Linked Debentures

Group No. 8

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INTRODUCTION:

In the evolving financial ecosystem, investments are majorly classified into two classes namely Equity and debt investments. Equity investments represent the ownership in a company thereby offering potentially high returns but involves higher ,market risks and volatility. Investors in equity instruments enjoy several benefits where most come through the dividends (a portion of company's profit) but they also face risk of huge losses during adverse market scenarios.

On the other hand, Debt investments is a type of investment which involves lending money to an entity/organization such as corporation or government in return for predetermined interest's payments throughout the period due to which they are generally considered less risky but they typically offer lower returns compared to equity investments due to very restricted exposure to their market growth. Common examples of debt investments include corporate bonds, Government bonds, etc.

Thus, due to an increase in demand by investors for a product that combines the safety of debt investments and higher returns like equity investments, Market -Linked Debentures have emerged as a new investment opportunity. MLD's are structured financial returns in which their returns are derived from the performance of an underlying market index such as equity indices ,government securities yields, etc. Unlike the regular bonds, MLD's typically pay the principal and the market -linked return in a single payout at the end of the tenure(maturity period).

Depending on their structure, they can be classified as:

Principal-Protected MLD's which guarantees the investor that is invested principle would be returned irrespective of the market conditions (performance of the underlying index) and the second type is Non- Principal-Protected MLD's where repayment of the principal is dependent on the performance of the underlying index .This type of structured investments allow the investors to enjoy the benefits from market linked returns but also manage the risk involved.

However, since the returns from MLD's are dependent on the performance of the underlying index, it is very important to quantify the risk and uncertainty in order to make the investor understand the type of gains, risk and losses they may face

Since, MLD's are affected by the unpredictable market movements, we need to develop an efficient and robust model which tells us how the returns behave and to make us understand how much money we can lose in different market conditions.

In this context , our study presents a parametric approach to model the return distribution for the three major stock indices- Nifty50, Banknifty and Sensex by using a variety of distributions which exhibit various characteristics such as skewness, asymmetry and heavy-tail . By using an egarch(1,1) model we try to capture asymmetric return behavior and volatility clustering . Further we try to identify the best fit probabilistic model for each index and estimate Value-at-Risk and Conditional tail expectation.

At the end ,we extend this risk modelling framework to Market-Linked Debentures by comparing their risk with direct equity investments. This is carried out by the use of both historical data as well as Monte-Carlo simulation for validating the accuracy of the model under consideration

LITERATURE REVIEW

Understanding the behavior of financial returns is necessary for accurately assessing the risk of structured investments such as Market-Linked Debentures. Traditional models that assume asset returns follow normal distribution often underestimate tail risk (risk of extreme events) because they fail to capture real-world patterns such as volatility clustering. To overcome these limitations, recent studies have employed flexible parametric distributions and filtered return models to eliminate volatility clustering to describe the return dynamics and improve the estimates obtained for Value-at-Risk and Conditional Tail Expectation.

Choi and Yoon (2020) introduced a comprehensive framework by applying 12 parametric distributions of different types such as Skewed, Symmetric and Fat-tailed to model global stock indices. Using various statistical tools for goodness of fit they found Generalized Hyperbolic and Variance-Gamma distributions provide better fits compared to normal and student's t models. They further extended this thought to Equity-linked Securities (ELS) and found that the heavy tailed distributions yield more accurate estimates of VaR and CTE. Their methodology informs the modelling strategy adopted in the current MLD study.

On the similar lines, Pokharel et al. (2024) compared Laplace-Family distributions and the Variance-Gamma distribution across different business sectors of S&P 500 and S&P 500 itself. The results produced from their research confirmed that both Laplace-family Distributions and Variance Gamma model effectively capture the asymmetry in the return data and the Variance-Gamma distribution exhibits the best performance in fat-tailed distributions.

In conclusion, all these studies provide a strong foundation for using parametric, heavy-tailed distributions for modelling financial risk. They demonstrate that fat-tailed and asymmetric models such as Generalized Hyperbolic, Skew t , Variance Gamma and Normal Inverse Gaussian capture the features of financial returns more effectively than those models with normality assumption.

The current research extends this idea by applying these distributions to Indian market indices – Nifty50, Banknifty and Sensex and further estimates the risk for Market-Linked Debentures using risk measures such as Value-at-Risk and Conditional Tail Expectation (CTE) and VaR backtesting, thereby extending the existing literature on structured-products risk which are emerging in the market.

OBJECTIVES:

1. To model the return distributional features of stock indices by fitting multiple parametric distributions including models comprising of symmetric, skewed and heavy-tailed forms in order to capture the financial characteristics observed in financial return data.
2. To identify the most optimal probabilistic model for each selected equity index.
3. To integrate the best-fit distributions into quantitative analysis focusing on the estimation of Value at Risk (VaR) and Conditional Tail Expectation (CTE) as measures of market risk.
4. To Evaluate the performance of risk models by conducting VaR backtesting and statistical coverage tests and thereby checking the accuracy of the model
5. To extend the modelling framework to Market Linked Debentures and thereby using the payoff formulas, carry risk estimation based on historical data and data generated from Monte-Carlo Simulation.
6. To interpret the risk associated with Market Linked Debentures in comparison with Equity Direct Investments

DATA PRE-PROCESSING:

In this study, we have used secondary data in the form of daily stock closing prices of Nifty 50, Banknifty, and Sensex obtained from the finance.yahoo.com website, over the past 10 years i.e. for the period Jan 02 2015 to August 30, 2025.

	Date	Adj Close	Close	High	Low	Open	Volume
1	2015-01-05	8378.40	8378.40	8445.60	8363.90	8407.95	118200
2	2015-01-06	8127.35	8127.35	8327.85	8111.35	8325.30	172800
3	2015-01-07	8102.10	8102.10	8151.20	8065.45	8118.65	164100
4	2015-01-08	8234.60	8234.60	8243.50	8167.30	8191.40	143800
5	2015-01-09	8284.50	8284.50	8303.30	8190.80	8285.45	148000
6	2015-01-12	8323.00	8323.00	8332.60	8245.60	8291.35	103200
7	2015-01-13	8299.40	8299.40	8356.65	8267.90	8346.15	129600
8	2015-01-14	8277.55	8277.55	8326.45	8236.65	8307.25	146100
9	2015-01-15	8494.15	8494.15	8527.10	8380.55	8424.50	232100
10	2015-01-16	8513.80	8513.80	8530.75	8452.25	8504.05	176000
11	2015-01-19	8550.70	8550.70	8570.95	8531.50	8550.05	125400

Return calculation: We need to model the distribution of the returns of the stocks therefore we first need to calculate returns. We calculated return of each stock closing price in R software itself by formula:

$$R_t = \log\left(\frac{P_t}{P_{t-1}}\right)$$

Where,

R_t : return value at time t

P_t : stock closing price at time t

P_{t-1} : stock closing price at time (t-1)

Data cleaning: After calculating the return, we checked whether there are any null values present in the data and removed that null values using R software.

METHODOLOGY:

1.Descriptive Statistics of Stock Returns:

A. Shapiro-wilk test

Used to check whether a dataset follows normal distribution

Ho: The data is normally distributed

VS

H1: Not Ho

Reject Ho if p-value<0.05

B. Jarque-bera test

It is a Goodness-of-fit test used to check whether a dataset follows normal distribution on the basis of its skewness and kurtosis

Ho: The data is normally distributed

VS

H1: Not Ho

Reject Ho if p-value<0.05

Table 1: Summary Statistics for the Three Stock Indices

Measures	Nifty50	Banknifty	Sensex
Obs.	2623	2347	2640
Mean	0.0004071647	0.0004410355	0.0003982771
Std. dev.	0.01042589	0.01510802	0.01044985
Skewness	-1.3507	-0.3595843	-1.334832
Kurtosis	20.0414	22.90221	21.01084
Jarque–Bera	< 2.2e-16	< 2.2e-16	< 2.2e-16
Shapiro wilk	< 2.2e-16	< 2.2e-16	< 2.2e-16

2. Time Series Analysis of stock returns

Figure 1-9 show the time series plot of daily returns for Nifty50,Banknifty and Sensex. We observe that all three indices volatility clustering meaning there are periods of high-volatility. Banknifty reflects greater market sensitivity since it shows higher volatility compared to the others.

The presence of fat -tails and asymmetry show that the return distribution is not normal, which indicates that we should consider heavy tailed parametric distributions.

A. For Nifty 50

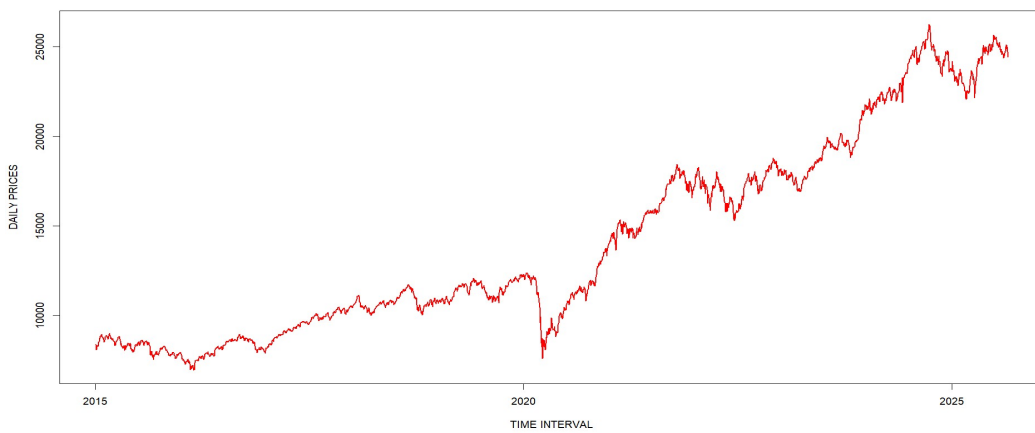


Fig1: Nifty50 index prices

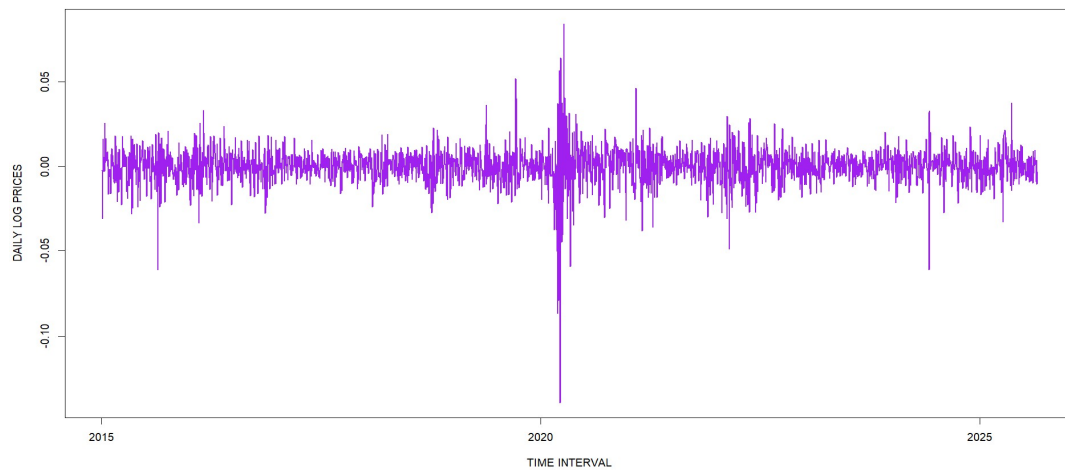


Fig2: Logarithmic time series of Nifty returns.

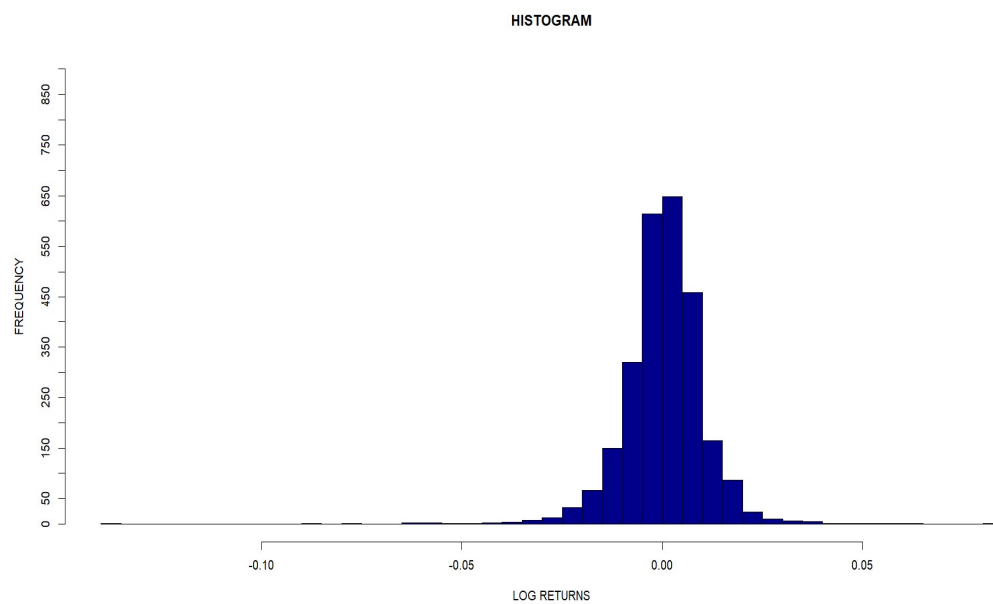


Fig3: Histogram of Nifty50 returns

B. For Banknifty

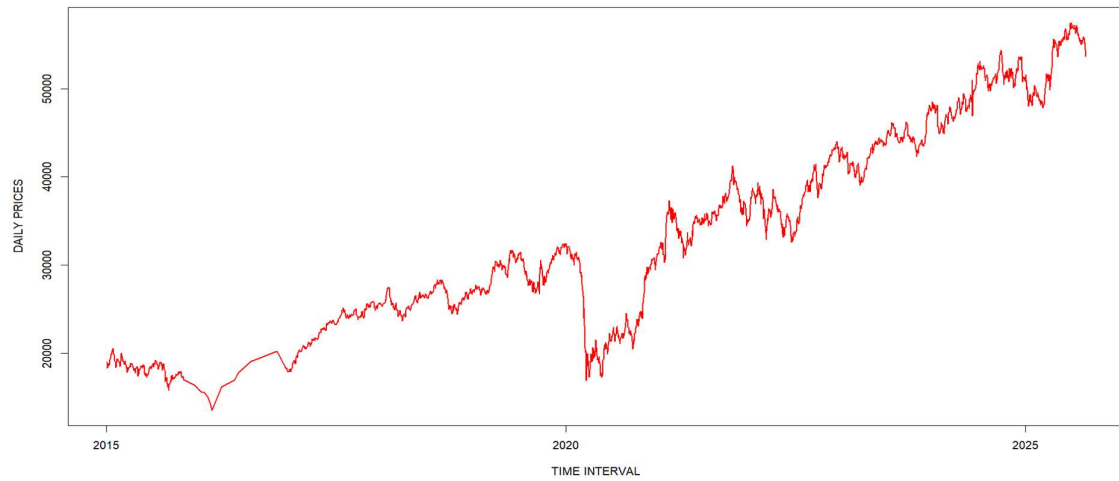


Fig4: Banknifty indexprices

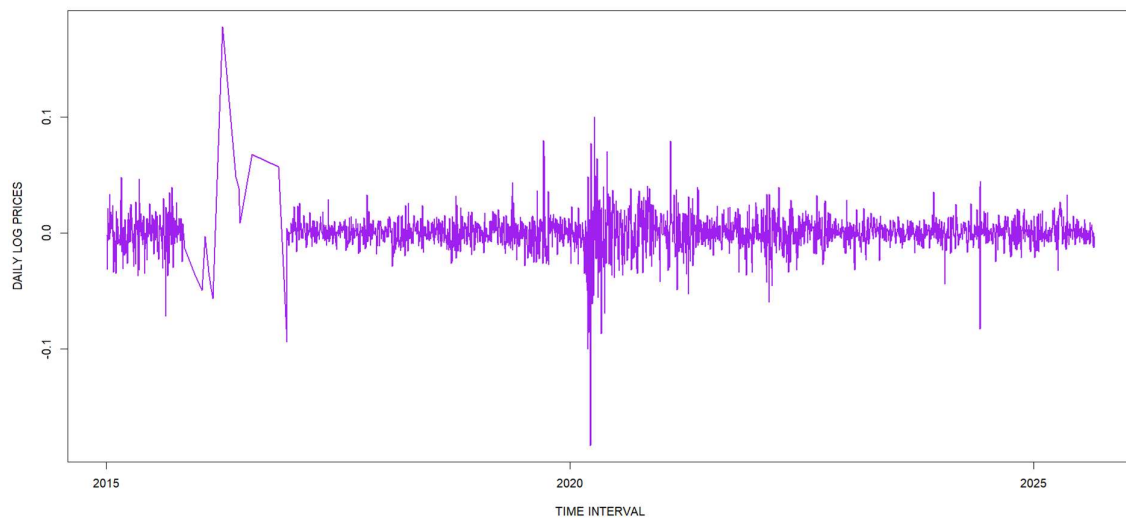


Fig5: Logarithmic time series of Banknifty returns.

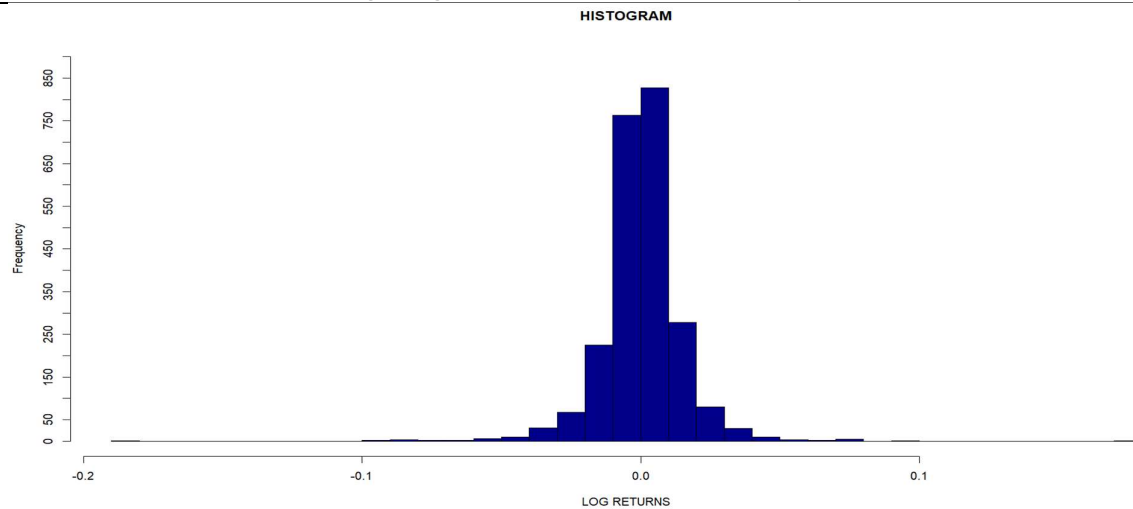


Fig5: Histogram of Banknifty returns

C. For Sensex

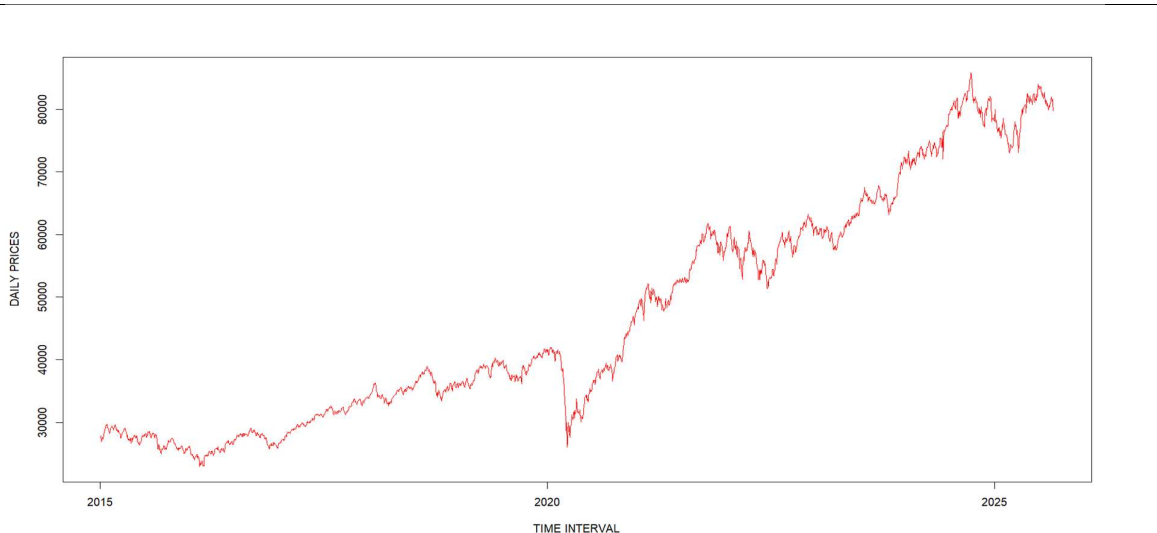


Fig7: Sensex index prices

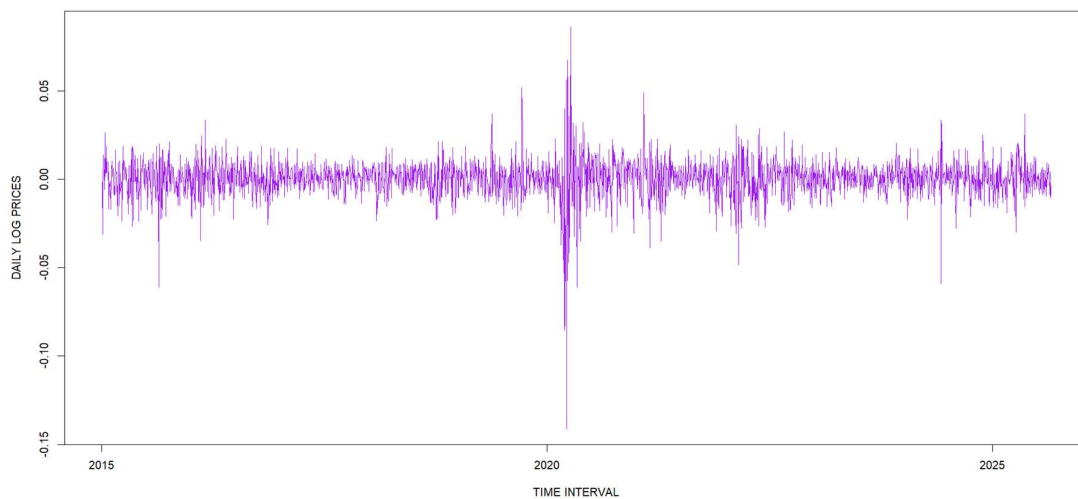


Fig8: Logarithmic time series of Sensex returns.

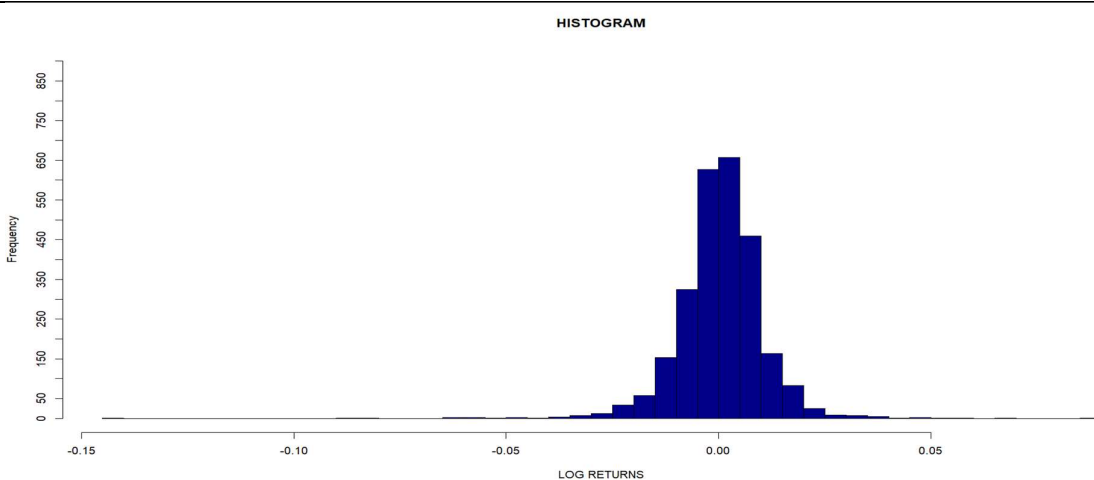


Fig9: Histogram of Sensex returns

AUTO CORRELATION FUNCTION:

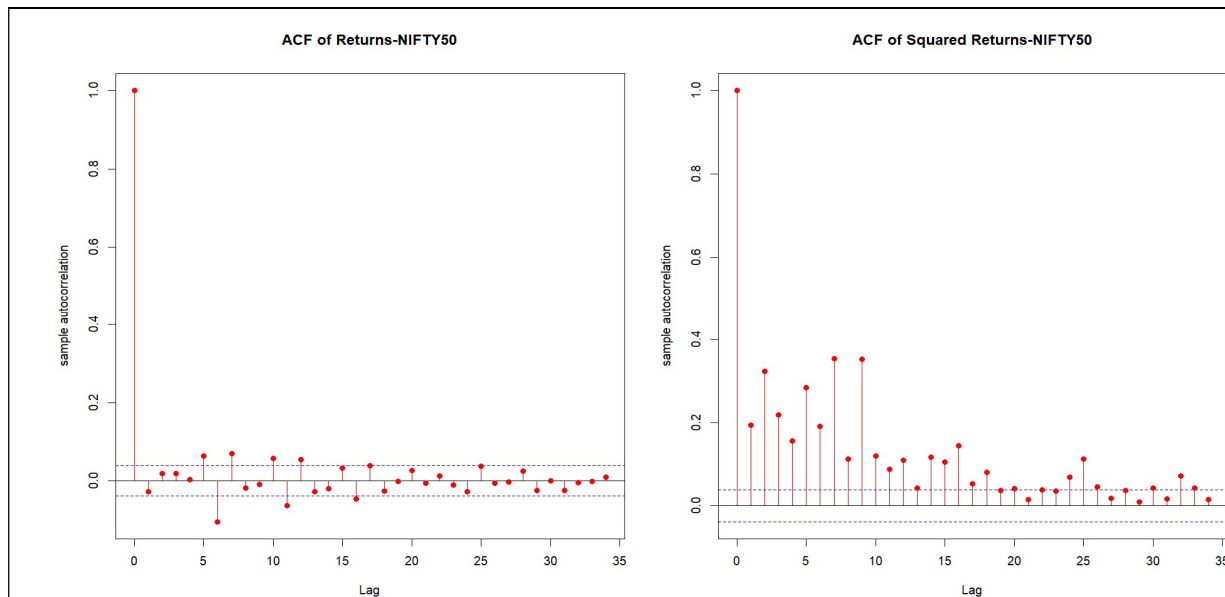


Fig 10. ACF of return and squared returns for Nifty50.

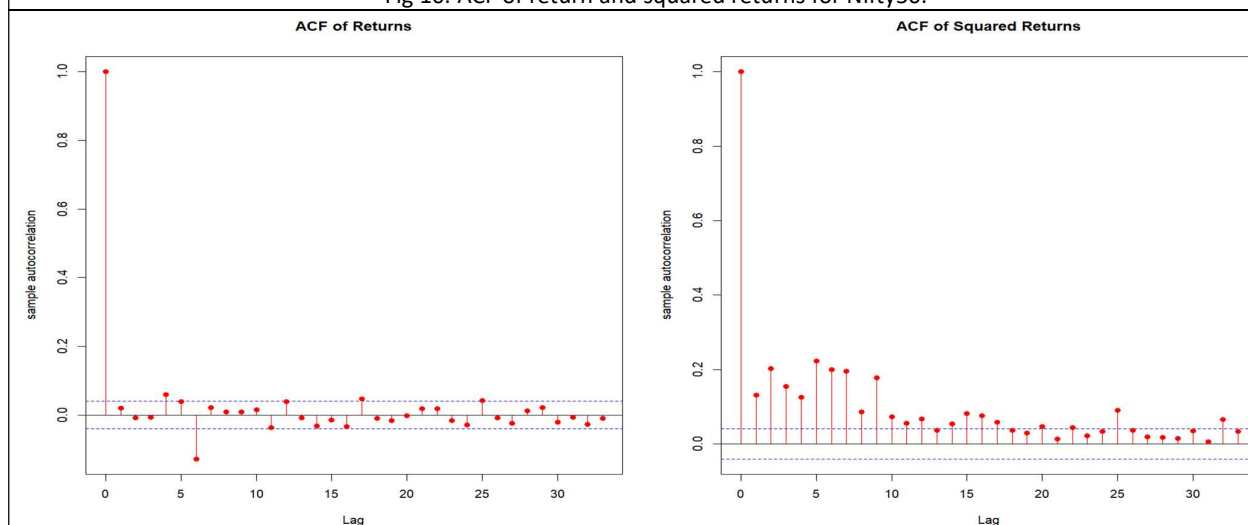


Fig 11: ACF of return and squared returns for Banknifty.

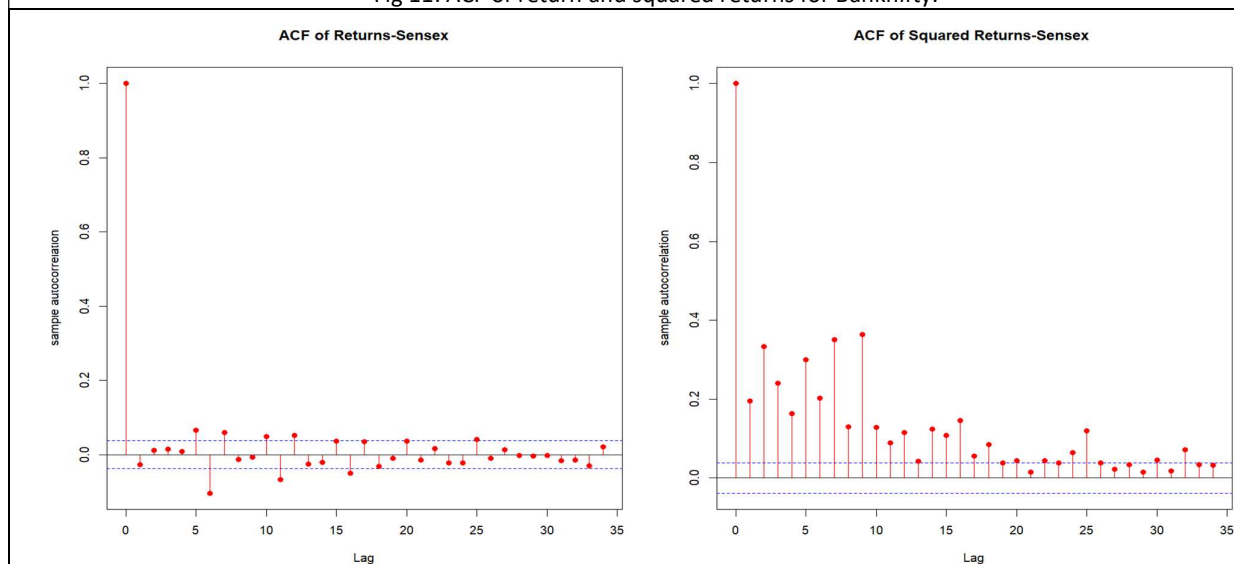


Fig12:ACF of return and squared returns for Sensex.

We can estimate the parameters of the distributions via method of maximum likelihood provided that the time series is approximately independent and identically distributed. When we plotted the ACF of the log return series and squared log return series in Figure 10-12 for all stock indices we found that the log returns are uncorrelated but squared log returns are highly correlated hence in this study we use an ARMA(1,1)-egarch(1,1) model under skew-t innovations in order to remove dependence among the series and get a time series which is approximately independent and identically distributed

The model was estimated using the rugarch package in R.

The specification is as follows:

$$r_t = \mu_t + \epsilon_t$$

Where,

r_t :return series

μ_t : conditional mean

$$\mu_t = C + \phi_1 r_{t-1} + \theta_1 \epsilon_{t-1}$$

Where,

C : constant term

ϕ_1 : The autoregressive AR(1) coefficient

θ_1 : The moving average MA(1) coefficient

$$\log(\sigma_t^2) = \omega + \alpha \left(\frac{|\epsilon_{t-1}|}{\sigma_{t-1}} - E\left[\frac{|\epsilon_{t-1}|}{\sigma_{t-1}}\right] \right) + \gamma \frac{\epsilon_{t-1}}{\sigma_{t-1}} + \beta \log(\sigma_{t-1}^2)$$

Where,

σ_t^2 : conditional variance

ω : constant term

α :The magnitude effect parameter which ,measures how volatility reacts to the magnitude caused due to past shocks

γ : The asymmetric (leverage) parameter.

β : The persistence parameter which represents the degree to which volatility remains over time

We create a filtered return series z_t as follows

$$z_t = \frac{\epsilon_t}{\sigma_t} = \frac{r_t - \mu_t}{\sigma_t}$$

z_t should be approximately independent and identically distributed.

The ACF of both filtered return series and squared filtered return series for each stock index is illustrated in Figure13-15 shows that there is no dependence in both filtered return series and squared return series

AUTO CORRELATION FUNCTION GRAPH FOR RESIDUALS OBTAINED FROM E-GARCH:

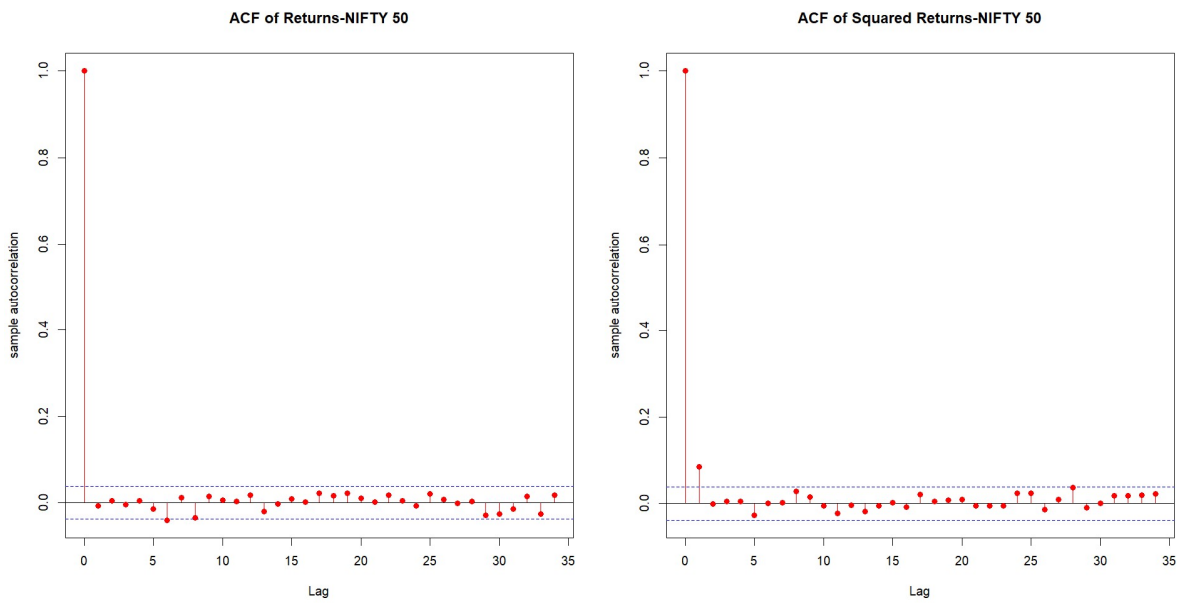


Fig 13. ACF of return and squared returns for the filtered Nifty50.

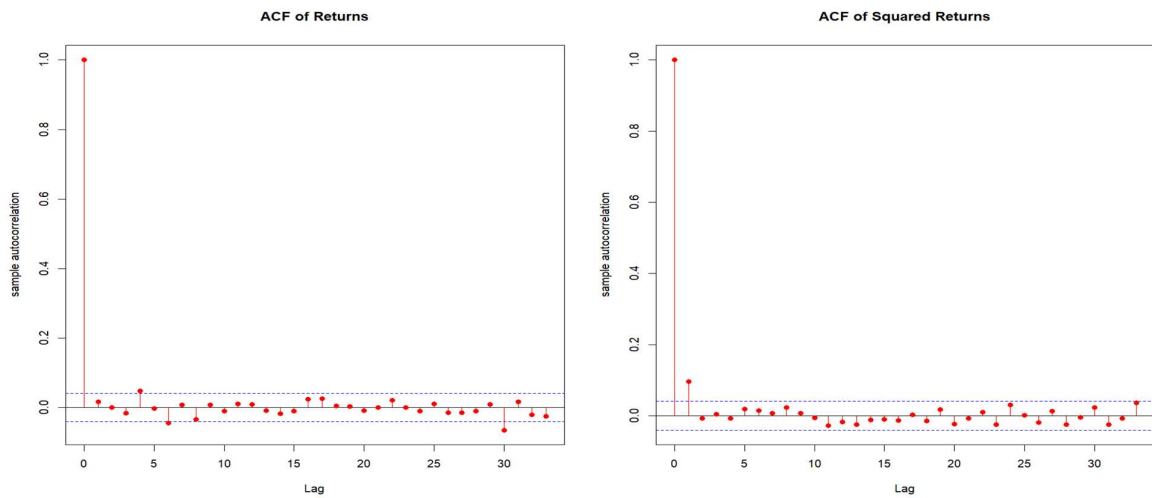


Fig 14: ACF of return and squared returns for the filtered Banknifty.

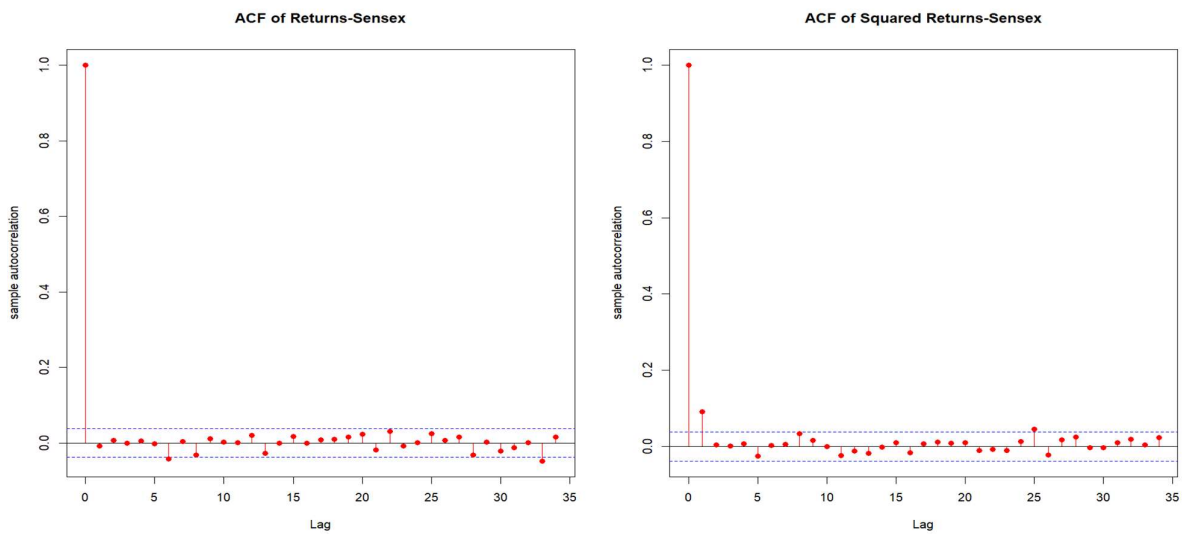


Fig15: ACF of return and squared returns for the filtered Sensex.

DISTRIBUTIONS UNDER CONSIDERATION:

In this study we have considered the following distributions- Cauchy, Laplace, normal, Student's t, skew normal, skew Cauchy, skew Laplace, skew Student's t, hyperbolic, normal inverse Gaussian, variance gamma, and generalized hyperbolic. because financial data typically show skewness ,heavy tails and excess kurtosis which cannot be captured by using Normal distribution for modelling. Hence more accurate modelling of return behavior and more reliable risk measures such as VaR and CTE can be analyzed and estimated by incorporating these alternative distributions such as Student's t, Skew t or generalized hyperbolic etc.

The MLE of these distributions can be estimated directly in R by using packages such as MASS, VGAM,sn,fGarch,ghyp.

1. Normal Distribution

PDF:

$$f(x; \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}.$$

Parameters:

μ : Location (mean); Range: $-\infty < \mu < \infty$

σ : Scale (standard deviation); Range: $\sigma > 0$

2. Laplace Distribution

PDF:

$$f(x; \mu, \sigma) = \frac{1}{2\sigma} \exp\left(-\frac{|x - \mu|}{\sigma}\right), \quad x \in \mathbb{R}.$$

Parameters:

μ : Location; Range: $-\infty < \mu < \infty$

σ : Scale; Range: $\sigma > 0$

.

3. Cauchy Distribution

PDF:

$$f(x; \mu, \sigma) = \frac{1}{\pi} \frac{\sigma}{\sigma^2 + (x - \mu)^2}, \quad x \in \mathbb{R}.$$

Parameters:

μ : Location; Range: $-\infty < \mu < \infty$

σ : Scale; Range: $\sigma > 0$

4. Student's t Distribution

PDF:

$$f(x; \mu, \sigma, \nu) = \frac{\Gamma(\nu + 1/2)}{\sqrt{\nu\pi}\sigma\Gamma(\nu/2)} \left(1 + \frac{1}{\nu} \left(\frac{x - \mu}{\sigma} \right)^2 \right)^{-(\nu+1/2)}, \quad x \in \mathbb{R}$$

Parameters:

μ : Location; Range: $-\infty < \mu < \infty$

σ : Scale; Range: $\sigma > 0$

ν : Degrees of freedom; Range: $\nu > 0$

5. Skew Normal Distribution

PDF:

$$f(x; \alpha) = 2\phi(x)\Phi(\alpha x), \quad x \in \mathbb{R}$$

Parameters:

μ : Location; Range: $-\infty < \mu < \infty$

σ : Scale; Range: $\sigma > 0$

α : Shape (skewness); Range: $-\infty < \alpha < \infty$

$\phi(x)$: Standard Normal PDF

$\Phi(x)$: Standard Normal CDF

6. Skew t Distribution

PDF:

$$f(x; \mu, \sigma, \alpha, \nu) = 2t(x; \mu, \sigma^2 + \alpha^2, \nu)T(\bar{x}; 0, 1, \nu + 1), \quad x \in \mathbb{R}$$

Parameters:

μ : Location; Range: $-\infty < \mu < \infty$

σ : Scale; Range: $\sigma > 0$

α : Shape (skewness); Range: $-\infty < \alpha < \infty$

ν : Degrees of freedom; Range: $\nu > 0$

$t(x; \mu, \sigma^2 + \alpha^2, \nu)$: PDF of a Student's t-distribution with location parameter μ , scale parameter $\nu(\sigma^2 + \alpha^2)$, and degrees of freedom ν .

$T(\bar{x}; 0, 1, \nu + 1)$: CDF of a Student's t-distribution with degrees of freedom $\nu + 1$ standardized mean 0, and variance 1

7. Skew Cauchy Distribution

PDF:

$$f(x; \mu, \sigma, \alpha) = \frac{\sigma [1 + 2 (\arctan(\alpha x / \sigma) / \pi)]}{\pi (\sigma^2 + x^2)}, \quad x \in \mathbb{R}$$

Parameters:

μ : Location; Range: $-\infty < \mu < \infty$

σ : Scale; Range: $\sigma > 0$

α : Shape (skewness); Range: $-\infty < \alpha < \infty$

8. Skew Laplace Distribution

PDF:

$$f(x; \mu, \sigma, \alpha) = \frac{1}{\sigma} \frac{\alpha}{1 + \alpha^2} \begin{cases} \exp\left(-\frac{\alpha}{\sigma}(x - \mu)\right), & \text{for } x \geq \mu, \\ \exp\left(\frac{1}{\alpha\sigma}(x - \mu)\right), & \text{for } x < \mu, \end{cases} \quad x \in \mathbb{R}$$

Parameters:

μ : Location; Range: $-\infty < \mu < \infty$

σ : Scale; Range: $\sigma > 0$

α : Shape (skewness); Range: $\alpha > 0$

9. Hyperbolic Distribution

PDF:

$$f(x; \mu, \sigma, \alpha, \beta) = \frac{\sqrt{\beta^2 - \alpha^2}}{2\beta\sigma K_1\left(\sigma\sqrt{\beta^2 - \alpha^2}\right)} \cdot \exp\left(-\beta\sqrt{\sigma^2 + (x - \mu)^2} + \alpha(x - \mu)\right), \quad x \in \mathbb{R}$$

Parameters:

μ : Location; Range: $-\infty < \mu < \infty$

σ : Scale; Range: $\sigma > 0$

α : Tail parameter; Range: $\alpha > |\beta|$

β : Skewness parameter; Range: $-\alpha < \beta < \alpha$

10. Normal Inverse Gaussian (NIG)

PDF:

$$f(x; \mu, \sigma, \alpha, \beta) = \frac{\sigma \beta \exp\left(\sigma \sqrt{\beta^2 - \alpha^2}\right) K_1\left(\beta \sqrt{\sigma^2 + (x - \mu)^2}\right) \exp(\alpha(x - \mu))}{\pi \sqrt{\sigma^2 + (x - \mu)^2}}, \quad x \in \mathbb{R}$$

Parameters:

μ : Location; Range: $-\infty < \mu < \infty$

σ : Scale; Range: $\sigma > 0$

α : Tail parameter; Range: $\alpha > |\beta|$

β : Skewness parameter; Range: $-\alpha < \beta < \alpha$

11. Variance Gamma (VG)

PDF:

$$f(x; \mu, \sigma, \alpha, \lambda) = \sqrt{\frac{\pi}{2}} \frac{\lambda^{1/\lambda} \exp((\alpha(x - \mu))/\sigma^2)}{\sigma \Gamma(1/\lambda)} \cdot \left(\frac{|x - \mu|}{\sqrt{\alpha^2 + (2\sigma^2/\lambda)}} \right)^{1/\lambda - 1/2} \cdot K_{1/\lambda - 1/2} \left(\frac{|x - \mu| \sqrt{\alpha^2 + (2\sigma^2/\lambda)}}{\sigma^2} \right), \quad x \in \mathbb{R}$$

Parameters:

μ : Location; Range: $-\infty < \mu < \infty$

σ : Scale; Range: $\sigma > 0$

α : Tail parameter; Range: $\alpha > |\beta|$

β : Skewness parameter; Range: $-\alpha < \beta < \alpha$

λ : Shape parameter; Range: $\lambda \in \mathbb{R}$

12. Generalized Hyperbolic (GHYP)

PDF:

$$f(x; \lambda, \mu, \sigma, \alpha, \beta) = \frac{(\beta^2 - \alpha^2) K_{\lambda-1/2}(\beta \sqrt{\sigma^2 + (x - \mu)^2}) \exp(\alpha(x - \mu))}{\sqrt{2\pi} \beta^{\lambda-1/2} \sigma^\lambda K_\lambda(\sigma \sqrt{\beta^2 - \alpha^2}) \left(\sqrt{\sigma^2 + (x - \mu)^2} \right)^{1/2-\lambda}},$$

Parameters:

σ : Scale; Range: $\sigma > 0$

α : Tail parameter; Range: $\alpha > |\beta|$

β : Skewness parameter; Range: $-\alpha < \beta < \alpha$

λ : Shape parameter; Range: $\lambda \in \mathbb{R}$

k_λ : a modified Bessel function of the third kind with index λ

ESTIMATION OF THE PARAMETERS FOR ALL 12 DISTRIBUTIONS BY METHOD OF MAXIMUM LIKELIHOOD:

Table 2 : Estimated parameters of the 12 distributions and maximum log-likelihood for the filtered residuals from Nifty 50 log-return:

Model	No.	Parameters	Log-Likelihood
Normal	2	$\mu = -0.001643094$ $\sigma = 1.008374884$	-3743.752
Laplace	2	$\mu = 0.02589474$ $\sigma = 0.7599479$	-3721.097
Cauchy	2	$\mu = 0.05333791$ $\sigma = 0.55351445$	-4020.997
Student's t	3	$\mu = 0.03191978$ $\sigma = 0.83779143$ $v = 6.66094094$	-3664.564
Skew Normal	3	$\mu = -0.008552$ $\sigma = 1.0023$ $\alpha = -0.2964$	-3706.151
Skew t	4	$\mu = 0.5503128$ $\sigma = 0.9710637$ $\alpha = -0.8188604$ $v = 7.1830738$	-3651.382
Skew Cauchy	3	$\mu = 0.1504727$ $\sigma = 0.5591503$ $\alpha = -0.1682965$	-4018.304
Skew Laplace	3	$\mu = 0.1202428$ $\sigma = 0.3768044$ $\alpha = 0.5396423$	-3715.666
Hyperbolic	4	$\mu = 0.3543193$ $\sigma = 0.9722877$ $\alpha = 2.1835993$ $\beta = -0.3559412$	-3656.839
NIG	4	$\mu = 0.3381640$ $\sigma = 0.9744375$ $\alpha = 2.3097136$ $\beta = -0.3398396$	-3654.764
Variance Gamma	4	$\mu = 0.3590726$ $\sigma = 0.9727949$ $\alpha = 2.9161088$ $\beta = -0.3609983$	-3660.434
Generalized Hyperbolic	5	$\mu = 0.3282408$ $\sigma = 0.9706962$ $\alpha = 0.3553765$ $\beta = -0.3317509$ $\lambda = -3.7582256$	-3651.330

Table 3: Estimated parameters of the 12 distributions and maximum log-likelihood for the filtered residuals from Banknifty log-return :

Model	No.	Parameters	Log-Likelihood
Normal	2	$\mu = 0.002138571$ $\sigma = 1.001612553$	-3334.030
Laplace	2	$\mu = 0.02518736$ $\sigma = -0.30152593$	-3266.135
Cauchy	2	$\mu = 0.02990741$ $\sigma = 0.52694983$	-3514.133
Student's t	3	$\mu = 0.02029246$ $\sigma = 0.78347466$ $v = 5.15857730$	-3231.475
Skew normal	3	$\mu = -0.002056149$ $\sigma = 0.998600513$ $\alpha = -0.161027361$	-3321.059
Skew t	4	$\mu = 0.002837675$ $\sigma = 1.000881561$ $\alpha = -0.293909414$ $v = 5.115471697$	-3229.426
Skew Cauchy	3	$\mu = 0.08021851$ $\sigma = 0.52794045$ $\alpha = -0.09261486$	-3513.340
Skew Laplace	3	$\mu = 0.07201351$ $\sigma = 0.3688099$ $\alpha = 0.5229751$	-3264.522
Hyperbolic	4	$\mu = 0.1104484$ $\sigma = 0.9840238$ $\alpha = 1.178022$ $\beta = -0.1084271$	-3234.988
NIG	4	$\mu = 0.1065187$ $\sigma = 0.989541$ $\alpha = 1.390647$ $\beta = -0.1037641$	-3231.930
Variance gamma	4	$\mu = 0.1136303$ $\sigma = 0.9833722$ $\alpha = 0$ $\beta = -0.1114982$	-3237.906
Generalized hyperbolic	5	$\mu = 0.1007881$ $\sigma = 0.9963472$ $\alpha = 1.106071$ $\beta = -0.09833108$ $\lambda = -1.838599$	-3230.176

Table 4 : Estimated parameters of the 12 distributions and maximum log-likelihood for the filtered residuals from Sensex log-return :

Model	No.	Parameters	Log- Likelihood
Normal	2	$\mu = -0.001550657$ $\sigma = 1.007949885$	-3766.902
Laplace	2	$\mu = 0.02855383$ $\sigma = -0.27428970$	-3745.784
Cauchy	2	$\mu = 0.04741825$ $\sigma = 0.55397388$	-4046.133
Student's t	3	$\mu = 0.02930778$ $\sigma = 0.83634164$ $v = 6.60694484$	-3687.248
Skew normal	3	$\mu = -0.00797037$ $\sigma = 1.00244156$ $\alpha = -0.27643297$	-3733.201
Skew t	4	$\mu = -0.0004277712$ $\sigma = 1.0101301057$ $\alpha = -0.4796498099$ $v = 2.9395900104$	-3676.817
Skew Cauchy	3	$\mu = 0.1339235$ $\sigma = 0.5583403$ $\alpha = -0.1490487$	-4044.035
Skew Laplace	3	$\mu = 0.1974131$ $\sigma = -0.1488065$ $\alpha = -0.2095389$	-3698.019
Hyperbolic	4	$\mu = 0.3153122$ $\sigma = 0.976661$ $\alpha = 2.132479$ $\beta = -0.3168815$	-3682.079
NIG	4	$\mu = 0.3015444$ $\sigma = 0.9788046$ $\alpha = 2.257476$ $\beta = -0.3031969$	-3679.897
Variance gamma	4	$\mu = 0.3169289$ $\sigma = 0.977206$ $\alpha = 0$ $\beta = -0.3184928$	-3685.643
Generalized hyperbolic	5	$\mu = 0.3150393$ $\sigma = 0.9972559$ $\alpha = 1.210704$ $\beta = -0.3296661$ $\lambda = -3.527633$	-3678.071

To Find the best fit distribution:

To identify the best fit distribution for each stock index we use several tools such as- Akaike Information Criterion(AIC), Bayesian Information Criterion(BIC), Kolmogorov-Smirnov test and graphical left tail tests .

1.AIC

$$AIC = 2 * K - 2\log L(\hat{\theta})$$

K : number of parameters in the model

$L(\hat{\theta})$: maximum likelihood value

2.BIC

$$BIC = K * \log(n) - 2\log L(\hat{\theta})$$

K : number of parameters in the model

$L(\hat{\theta})$: maximum likelihood value

n : number of observations

3.KOLMOGOROV-SMIRNOV TEST:

The K-S test is a non-parametric goodness of fit test used to determine whether a sample follows a specified theoretical distribution

Ho: The data follows the specified distribution

Vs

H1: The data does not follow the specified distribution

The test-statistic is

$$D = \sup |F_n(x) - F(x)|$$

Reject Ho if $D > D_{table}$ or p-value $< \alpha$

4.GRAPHICAL TEST

In addition to the information criteria (AIC and BIC) and the Kolmogorov–Smirnov goodness-of-fit test, a graphical tail analysis was performed to visually assess the adequacy of the fitted distributions, particularly in the extreme regions of the return distribution. The graphical tail test involves plotting the empirical cumulative distribution function (CDF) of the observed data against the theoretical CDF of each fitted model on a logarithmic scale, focusing on the left-hand (loss) tail.

PROCEDURE:

1. The order statistics of the empirical return series were computed and plotted against the corresponding theoretical probabilities from each fitted distribution.
2. The logarithm of these cumulative probabilities was taken to emphasize tail behavior.
3. Each of the fitted distributions was overlaid on the same plot for direct comparison.

The results are as follows:

Table 5: (AIC,BIC) values of the 12 distributions from – Sensex, Banknifty, Nifty50.

Model	Nifty50	Banknifty	Sensex
Normal	(7491.503, 7503.247)	(6672.061, 6683.582)	(7537.805, 7549.562)
Laplace	(7446.194, 7457.938)	(6536.269, 6547.791)	(7495.567, 7507.325)
Cauchy	(8045.994, 8057.738)	(7032.266, 7043.788)	(8096.265, 8108.022)
Student's t	(7335.129, 7352.745)	(6468.950, 6486.233)	(7380.496, 7398.131)
Skew normal	(7418.301, 7435.918)	(6648.119, 6665.402)	(7472.401, 7490.037)
Skew t	(7310.764, 7334.252)	(6466.851, 6489.895)	(7361.634, 7385.148)
Skew Cauchy	(8042.607, 8060.224)	(7032.680, 7049.962)	(8094.069, 8088.069)
Skew Laplace	(7437.333, 7454.949)	(6535.045, 6552.327)	(7404.038, 7427.552)
Hyperbolic	(7321.678, 7345.167)	(6477.976, 6501.020)	(7372.157, 7403.550)
NIG	(7317.529, 7341.017)	(6471.860, 6494.903)	(7367.794, 7399.187)
Variance gamma	(7328.868, 7352.357)	(6483.812, 6506.856)	(7379.286, 7410.679)
Generalized hyperbolic	(7312.661, 7342.021)	(6470.352, 6499.157)	(7366.141, 7395.534)

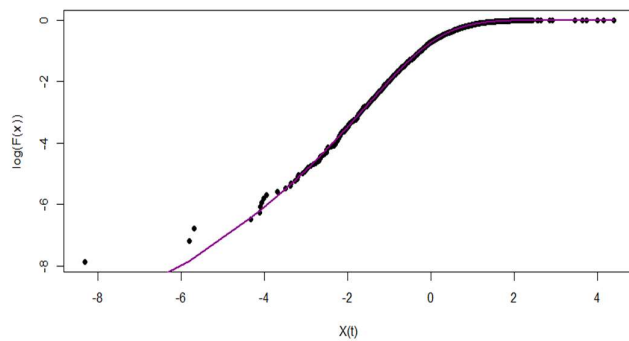
RESULTS OF K-S TEST

Table 6: K-S test statistic and p-value for-Nifty50,Banknifty and Sensex

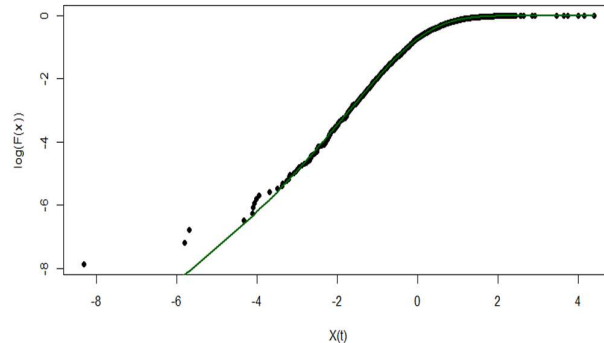
Model	Nifty50	Banknifty	Sensex
Normal Statistic p value	0.0481 ≤0.0000	0.04524 ≤0.0000	0.04813 ≤0.0000
Laplace Statistic p value	0.0475 ≤0.0000	0.03446 ≤0.0000	0.04040 ≤0.0000
Cauchy Statistic p value	0.0736 ≤0.0000	0.06364 ≤0.0000	0.07331 ≤0.0000
Student's t Statistic p value	0.0179 0.36552	0.00826 0.99712	0.04790 ≤0.0000
Skew normal Statistic p value	0.03198 0.00937	0.040999941 ≤0.0000	0.02858 ≤0.0000
Skew t Statistic p value	0.01670 0.45712	0.00846 0.9960	0.01698 0.43161
Skew Cauchy Statistic p value	0.0661 ≤0.0000	0.06118 ≤0.0000	0.06506 ≤0.0000
Skew Laplace Statistic p value	0.0481 ≤0.0000	0.03258 ≤0.0000	0.02314 0.11807
Hyperbolic Statistic p value	0.0177 0.3810	0.0107 0.9498	0.01763 0.38456
Normal inverse Gaussian Statistic p value	0.0173 0.4104	0.0110 0.9372	0.01726 0.41052
Variance gamma Statistic p value	0.0186 0.3214	0.0120 0.8874	0.01869 0.31465
Generalized hyperbolic Statistic p value	0.0156 0.5489	0.0080 0.9979	0.01721 0.41447

RESULTS OF GRAPHICAL LEFT TAIL TEST: For Nifty 50:

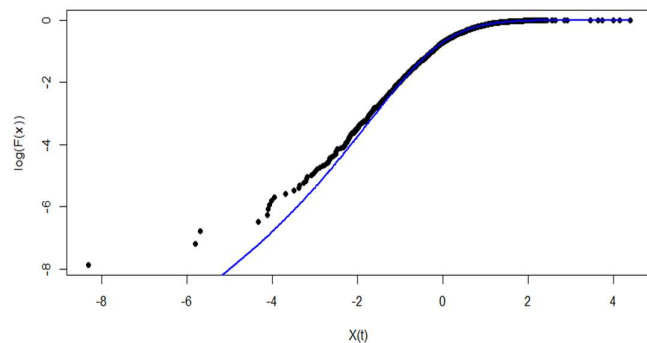
Graphical Tail Test - Generalized Hyperbolic



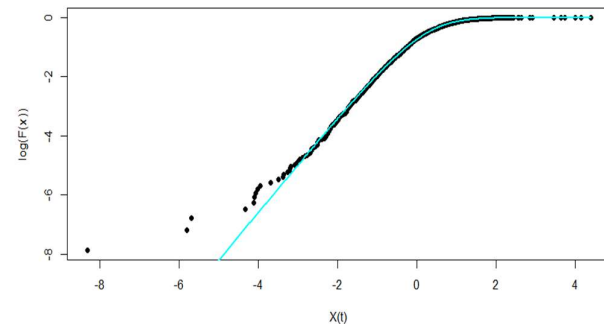
Graphical Tail Test - Skew t



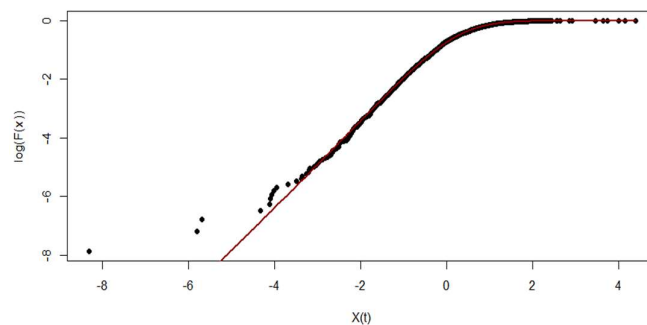
Graphical Tail Test - Student's t



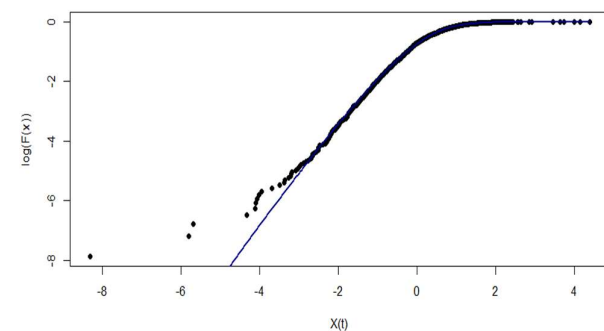
Graphical Tail Test - Hyperbolic



Graphical Tail Test - Normal Inverse Gaussian

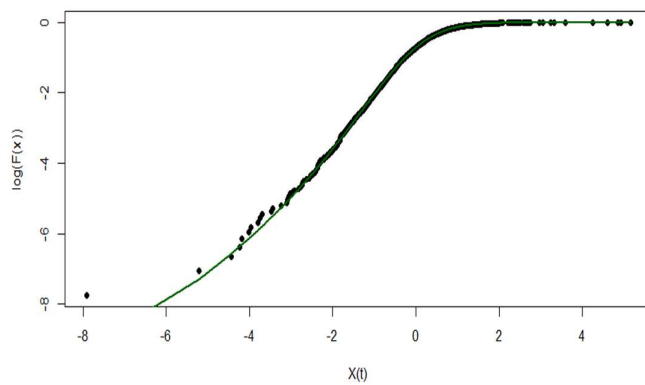


Graphical Tail Test - Variance Gamma

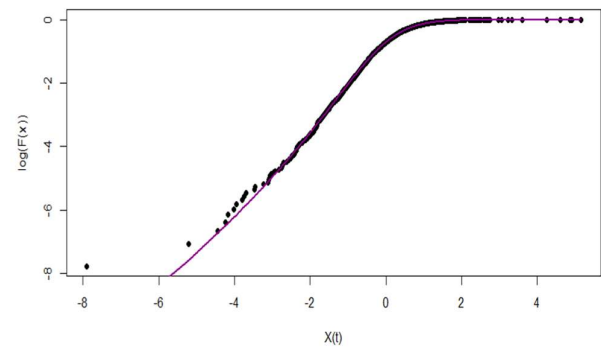


For Banknifty:

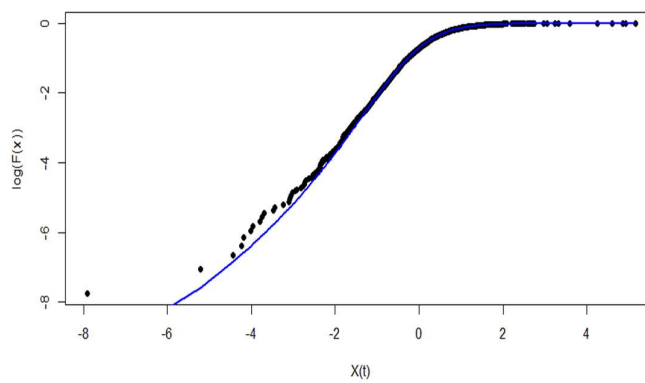
Graphical Tail Test - Skew t



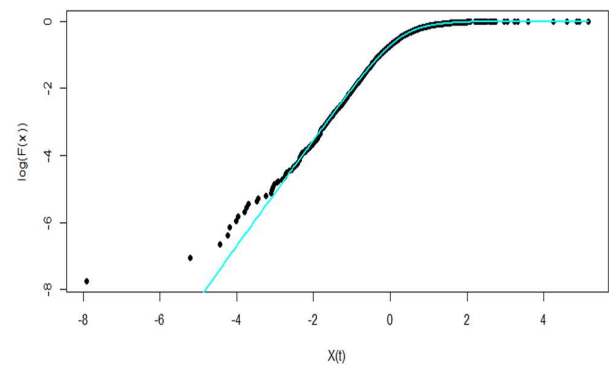
Graphical Tail Test - Generalized Hyperbolic



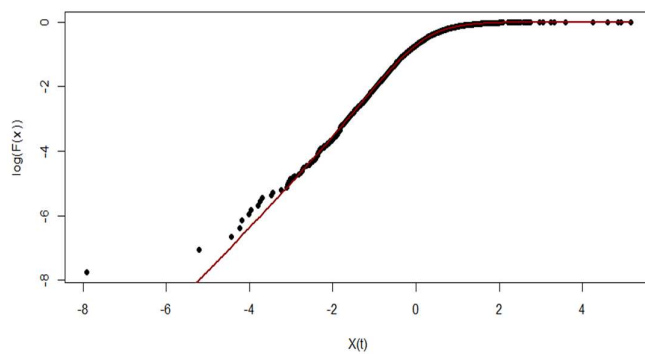
Graphical Tail Test - Student's t



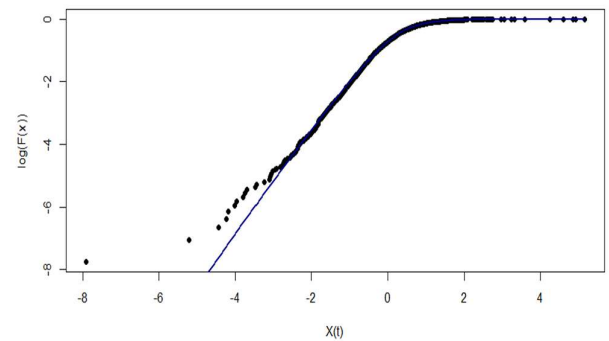
Graphical Tail Test - Hyperbolic



Graphical Tail Test - Normal Inverse Gaussian

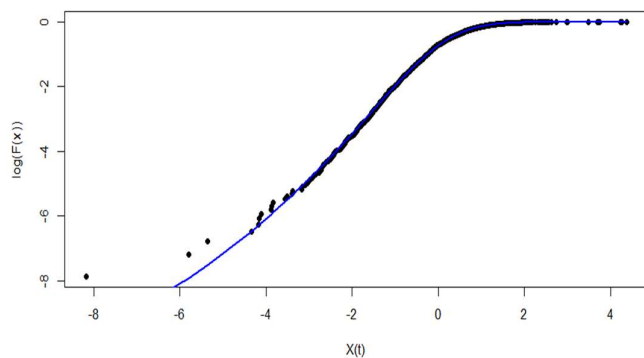


Graphical Tail Test - Variance Gamma

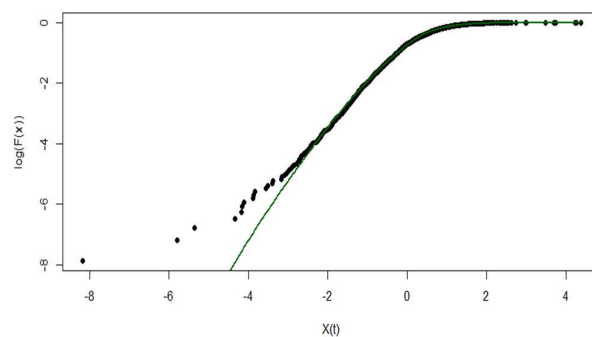


For Sensex:

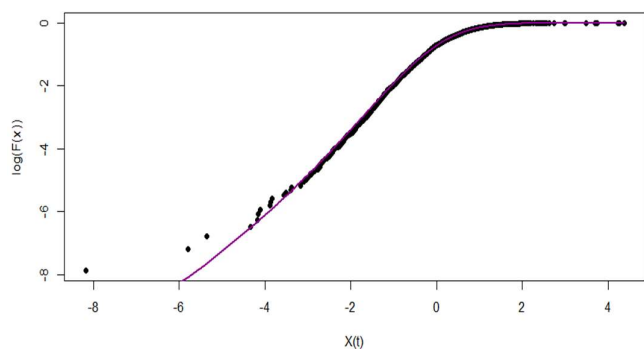
Graphical Tail Test - Skew t



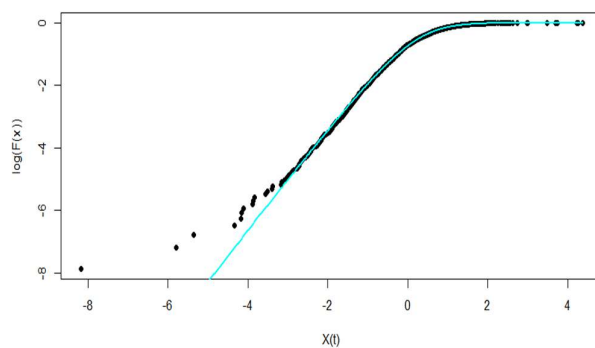
Graphical Tail Test - Skew laplace



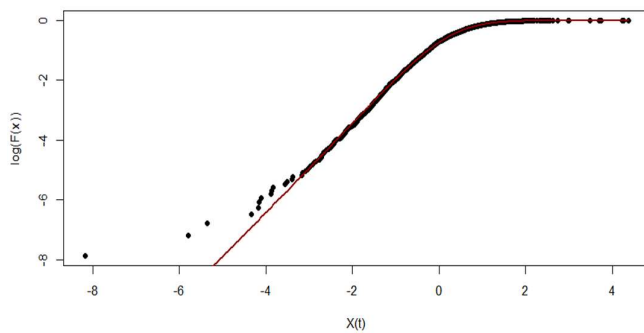
Graphical Tail Test - Generalized Hyperbolic



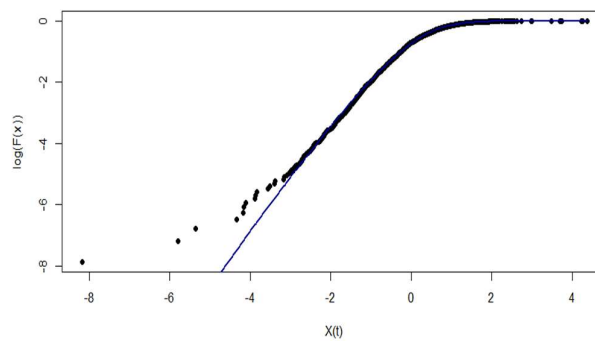
Graphical Tail Test - Hyperbolic



Graphical Tail Test - Normal Inverse Gaussian



Graphical Tail Test - Variance Gamma



Therefore ,by combining the results obtained from all tools ,the best fit distribution for each stock index are as follows:

Table 7:Best Fit Distributions

Stock index	Best fit distribution
Nifty50	Generalized Hyperbolic
Banknifty	Skew-t
Sensex	Skew-t

Risk Analysis

1. VALUE AT RISK:

Value at Risk (VaR) is a statistical measure that estimates the maximum expected loss on an investment over a specified time period at given time interval. The VaR shows how much a portfolio loses within a certain time period.

VaR at a confidence level $\theta \in (0,1)$ for Loss L of a security portfolio is defined to be

$$VaR = \inf \{l \in \mathbb{R} : F_L(l) \geq \theta\}$$

Where F is the distribution of loss function

2. CONDITIONAL TAIL EXPECTATION:

Conditional Tail expectation (CTE) measures the average loss that occurs beyond the Value at Risk (VaR) level i.e. it tells the expected loss in worst case scenario. It indicates the expected loss when the loss occurred loss is greater than VaR.

CTE at a confidence level $\theta \in (0,1)$ for Loss L of a security portfolio is defined to be

$$CTE_{\theta} = E\{L|L < VaR_{\theta}\}$$

APPLICATION TO MARKET-LINKED DEBENTURES:

There are several kinds of mld of which we have considered Principal-Protected and Non-Principal protected for our study

The Payoff and Losses for each structure were defined as :

1. Principal-Protected MLD:

$$Payoff = Principal (1 + \max(0, Prate * returns))$$

2. Non-Principal-Protected MLD:

$$Payoff = Principal (1 + Prate * returns)$$

$$Loss = \frac{Principal - Payoff}{Principal}$$

Where

p_{rate} :The participation rate determines how much of the underlying asset's return is passed on to the investor in the structured product.

For Direct Investments,

$$Loss = -returns$$

In this study

$Principal = Rs\ 100$

$n = maturity\ period = 3\ years$

$Prate = 80\%$

Table 9 : Risk measures for the equally weighted portfolio(direct investment) via historical data

Measures	Nifty50 (%)	Banknifty(%)	Sensex(%)
VaR(99%)	2.701793	4.182023	2.699757
VaR(95%)	1.497767	2.113891	1.474738
CTE(99%)	4.572059	6.689879	4.606388
CTE(95%)	2.454243	3.591946	2.452327

Table 10 : Risk measures for the principal-Protected Market-linked debentures

Measures	Nifty50(%)	Banknifty(%)	Sensex(%)
VaR(99%)	0	0	0
VaR(95%)	0	0	0
CTE(99%)	0	0	0
CTE(95%)	0	0	0

Table 11 : Risk measures for the Non- Principal-Protected Market-linked debentures

Measures	Nifty50(%)	Banknifty(%)	Sensex(%)
VaR(99%)	2.161434	3.345618	2.159806
VaR(95%)	1.198213	1.691113	1.179791
CTE(99%)	3.657648	5.351903	3.685110
CTE(95%)	1.963395	2.873557	1.961861

APPLICATION TO STOCK INDEX:

In this section, we implement backtest for VaR . We compared the daily VaR estimates at the 99% Confidence level. Backtesting means checking your model's performance using past data to see whether it would have worked as expected before implementing it in future. In order to forecast one day ahead VaR we used all the observations

The one-day ahead VaR is given as:

$$VaR_{t+1}(\hat{\theta}) = \hat{\mu}_{t+1} + q_{\theta} * \hat{\sigma}_{t+1}$$

Where ,

q_{θ} = quantile at the given probability

$\hat{\mu}_{t+1}$ = forecasted value of mean by egarch model

$\hat{\sigma}_{t+1}$ = forecasted value of mean by egarch model

For each day, compare the actual return (or loss) L_t and the Predicted (VaR_t)

If $L_t < -VaR_t$ it is called as violation or exception meaning the actual loss is greater than predicted loss

A violation of the VaR model is defined by

$$X_{VaR}(\theta) = \sum_{i=1}^N 1_{\{L_i < -VaR(\theta)\}}$$

For this purpose , we will use an unconditional conditional coverage test and conditional coverage test.

1.UNCONDITIONAL COVERAGE TEST(LRuc) :

It is used to check whether the actual number of exceedances matches the expected number as predicted by the model

Ho: The actual proportion of VaR exceedances equals the expected proportion

Vs

H1: The actual proportion of VaR exceedances differs from the expected proportion

N =Total number of observations

x = actual number of exceedances

$\hat{p}$ = expected exceedance probability

$$LR_{uc} = -2 \log \frac{(1-p)^{N-x} p^x}{(1-\hat{p})^{N-x} \hat{p}^x}$$

Where,

$$\hat{p} = \frac{x}{N}$$

$$LR_{uc} \sim \chi^2(1)$$

Decision criteria: Reject Ho if $LR_{uc} > \chi^2(1, 1 - \alpha)$ or p-value $> \alpha$

2. CHRISTOFFERSEN CONDITIONAL COVERAGE TEST (LR_{cc}):

It is used to check if exceptions occur independently over time and not clustered together.

Ho: The model provides correct coverage and the exceptions are independent over time.

VS

H1: Either the model does not have correct coverage, or the exceptions are not independent means they are clustered.

$$LR_{cc} = LR_{uc} + LR_{ind}$$

Where ,

$$LR_{ind} = -2\ln\left[\frac{(1 - \hat{\pi})^{n_{00}+n_{10}} (\hat{\pi})^{n_{01}+n_{11}}}{(1 - \hat{\pi}_0)^{n_{00}} (\hat{\pi}_0)^{n_{01}} (1 - \hat{\pi}_1)^{n_{10}} (\hat{\pi}_1)^{n_{11}}}\right]$$

$$LR_{cc} \sim \chi^2(2)$$

Decision criteria: Reject Ho if $LR_{cc} > \chi^2(2, 1 - \alpha)$ or p-value $> \alpha$

The results for VaR backtesting were as follows:

Table 8:

Tests	Nifty50	Banknifty	Sensex
Expected Violations	26	23	26
Actual Violations ($N * 0.01$)	24	22	24
Kupiec unconditional Coverage test Statistic	0.1971162	0.09498533	0.2273146
p value	0.6570587	0.7579325	0.6335222
Conditional Coverage test Statistic	0.6405406	0.5115148	0.6678561
p value	0.7259528	0.7743298	0.7161053
Decision criteria	Do not Reject Ho ,for both test	Do not Reject Ho ,for both test	Do not Reject Ho ,for both test

The backtesting results confirm that the VaR model performs well across all three indices. Hence, the model is considered statistically reliable and efficient for measuring market risk in these indices

RISK ANALYSIS BY USING MONTE CARLO SIMULATION:

Using the best-fit Parametric distribution we generate simulated returns with 100000 draws and then carried out the same procedure as done for calculation of VaR and CTE as done previously for historical data and the results were as follows:

Table 12 : Risk measures for the equally weighted portfolio(direct investment) via simulation

Measures	Nifty50 (%)	Banknifty(%)	Sensex(%)
VaR(99%)	2.827698	2.682138	1.293541
VaR(95%)	1.663599	1.597946	0.7632070
CTE(99%)	3.726811	3.521513	1.673470
CTE(95%)	2.413866	2.293143	1.1037623

Table 13 : Risk measures for the principal-Protected Market-linked debenture via simulation

Measures	Nifty50(%)	Banknifty(%)	Sensex(%)
VaR(99%)	0	0	0
VaR(95%)	0	0	0
CTE(99%)	0	0	0
CTE(95%)	0	0	0

Table 14 : Risk measures for the Non- Principal-Protected Market-linked debenture via simulation

Measures	Nifty50(%)	Banknifty(%)	Sensex(%)
VaR(99%)	2.262158	2.145711	1.034833
VaR(95%)	1.330879	1.278357	0.6105656
CTE(99%)	2.981449	2.817211	1.338776
CTE(95%)	1.931093	1.834515	0.8830098

**RESULTS AND DISCUSSION:
USING HISTORICAL DATA
FOR DIRECT INVESTMENTS,**

1. At 99% confidence interval VaR for Nifty50 was approximately 2.7% and CTE stood at 4.57% Indicating that the expected average loss could reach around 4.57% on worst 1% trading days
2. At 95% confidence interval VaR for Nifty50 was approximately 1.49% and CTE stood at 2.45% Indicating that the expected average loss could reach around 2.45% on worst 5% trading days.
3. At 99% confidence interval VaR for Banknifty was approximately 4.18% and CTE stood at 6.68% Indicating that the expected average loss could reach around 6.68% on worst 1% trading days
4. At 95% confidence interval VaR for Banknifty was approximately 1.69% and CTE stood at 2.87% Indicating that the expected average loss could reach around 2.87% on worst 5% trading days.
5. At 99% confidence interval VaR for Sensex was approximately 2.69% and CTE stood at 4.6% Indicating that the expected average loss could reach around 4.6% on worst 1% trading days
6. At 95% confidence interval VaR for Sensex was approximately 1.47% and CTE stood at 2.45% Indicating that the expected average loss could reach around 2.45% on worst 5% trading days.

FOR INDIRECT INVESTMENTS:

A.PRINCIPAL-PROTECTED MLD:

For Principal-Protected MLD, the VaR and CTE is 0% across for all indices. This reflects the feature of capital protection of the product that ensures the consumer does not lose their initial investment under adverse market conditions

B.NON- PRINCIPAL-PROTECTED MLD:

1. At 99% confidence interval VaR for MLD linked to Nifty50 was approximately 2.7% and CTE stood at 4.57% Indicating that the expected average loss could reach around 4.57% on worst 1% trading days
2. At 95% confidence interval VaR for MLD linked Nifty50 was approximately 1.49% and CTE stood at 4.57% Indicating that the expected average loss could reach around 2.45% on worst 5% trading days.

3. At 99% confidence interval VaR for MLD linked to Banknifty was approximately 3.34% and CTE stood at 5.35% Indicating that the expected average loss could reach around 5.35% on worst 1% trading days
4. At 95% confidence interval VaR for MLD linked Banknifty was approximately 1.69% and CTE stood at 2.87% Indicating that the expected average loss could reach around 2.87% on worst 5% trading days.
5. At 99% confidence interval VaR for MLD linked to Sensex was approximately 2.15% and CTE stood at 3.68% Indicating that the expected average loss could reach around 3.68% on worst 1% trading days
6. At 95% confidence interval VaR for MLD linked to Sensex was approximately 1.17% and CTE stood at 1.96% Indicating that the expected average loss could reach around 1.96% on worst 5% trading days.

**USING SIMULATED DATA
FOR DIRECT INVESTMENTS,**

1. At 99% confidence interval VaR for Nifty50 was approximately 2.82% and CTE stood at 3.72% Indicating that the expected average loss could reach around 3.72% on worst 1% trading days
2. At 95% confidence interval VaR for Nifty50 was approximately 1.66% and CTE stood at 2.41% Indicating that the expected average loss could reach around 2.41% on worst 5% trading days.
3. At 99% confidence interval VaR for Banknifty was approximately 2.68% and CTE stood at 3.52% Indicating that the expected average loss could reach around 3.52% on worst 1% trading days
4. At 95% confidence interval VaR for Banknifty was approximately 1.59% and CTE stood at 2.29% Indicating that the expected average loss could reach around 2.29% on worst 5% trading days.
5. At 99% confidence interval VaR for Sensex was approximately 1.29% and CTE stood at 1.67% Indicating that the expected average loss could reach around 1.67% on worst 1% trading days
6. At 95% confidence interval VaR for Sensex was approximately 0.76% and CTE stood at 1.1% Indicating that the expected average loss could reach around 1.1% on worst 5% trading days.

FOR INDIRECT INVESTMENTS:**A. PRINCIPAL-PROTECTED MLD:**

1. For Principal-Protected MLD, the VaR and CTE is 0% across for all indices. This reflects the feature of capital protection of the product that ensures the consumer does not lose their initial investment under adverse market conditions

B. NON- PRINCIPAL-PROTECTED MLD:

1. At 99% confidence interval VaR for MLD linked to Nifty50 was approximately 2.62% and CTE stood at 2.98% Indicating that the expected average loss could reach around 2.98% on worst 1% trading days
2. At 95% confidence interval VaR for MLD linked Nifty50 was approximately 1.33% and CTE stood at 1.93% Indicating that the expected average loss could reach around 1.93% on worst 5% trading days.
3. At 99% confidence interval VaR for MLD linked to Banknifty was approximately 2.14% and CTE stood at 2.81% Indicating that the expected average loss could reach around 2.81% on worst 1% trading days
4. At 95% confidence interval VaR for MLD linked Banknifty was approximately 1.27% and CTE stood at 1.83% Indicating that the expected average loss could reach around 1.83% on worst 5% trading days.
5. At 99% confidence interval VaR for MLD linked to Sensex was approximately 1.03% and CTE stood at 1.33% Indicating that the expected average loss could reach around 3.68% on worst 1% trading days.
7. At 95% confidence interval VaR for MLD linked to Sensex was approximately 0.61% and CTE stood at 0.88% Indicating that the expected average loss could reach around 0.88% on worst 5% trading days.

CONCLUSIONS

1. The return distributions of each stock index exhibit asymmetry and fat tails indicating that normal distribution is unsuitable for modelling Indian equity return behavior.
2. Using VaR Backtesting and statistical coverage tests, the models were confirmed to produce tail risks projections.
3. For Direct investments, using both Empirical(historical data) and Monte Carlo Simulation ,it was found that Banknifty exhibits the highest risk followed by Nifty50 and Sensex.
4. For Market-Linked Debentures,
 - a. For Principal-Protected MLDs , both VaR and CTE were zero reflecting the feature complete capital protection in all scenarios.
 - b. Non-Principal-Protected MLDs ,showed moderate risk-levels, higher than Principal-Protected MLDs and lower than Direct-Equity investments .
 - c. Direct-Equity investments were found to be riskier because of complete exposure to market fluctuations especially in adverse market scenarios.

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3. GARCH(1,1) models
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<https://www.incredmoney.com/blog/an-introduction-to-market-linked-debentures/>

ANNEXURE**For Nifty 50****For egarch**

```
library(rugarch)
```

```
library(PerformanceAnalytics)
```

```
spec_egarch <- ugarchspec(
  variance.model = list(model = "eGARCH", garchOrder = c(1,1)),
  mean.model     = list(armaOrder = c(1,1), include.mean = TRUE),
  distribution.model = "sstd" # skewed Student-t innovations
)
```

```
fit_egarch <- ugarchfit(spec_egarch, data = nifty$returns)
show(fit_egarch)
f1=residuals(fit_egarch, standardize = TRUE)
library(xts)
f= xts(f1, order.by = as.Date(nifty$Date))
```

For VaR Backtesting

```
alpha <- 0.01 # 99% VaR level
q_ghyp <- qghyp(alpha, object = fit_ghyp)
mu_t <- fitted(fit_egarch) # conditional mean from EGARCH
sigma_t <- sigma(fit_egarch) # conditional SD from EGARCH
```

```
VaR_99 <- mu_t + sigma_t * q_ghyp; VaR_99
violations <- ifelse(nifty$returns < VaR_99, 1, 0)
sum(violations) # number of violations
```

```
length(violations) # total observations
#install.packages("VaRTest")
#library(VaRTest)
library(PerformanceAnalytics)
alpha <- 0.01
kupiec <- VaRTest(alpha, actual = nifty$returns, VaR = VaR_99)
print(kupiec)
```

Calculation of VaR and CTE:

```
loss=-nifty$returns
alpha <- 0.01
VaR_emp99 <- quantile(loss, 1-alpha)
CTE_emp99 <- mean(loss[loss> VaR_emp99])
VaR_emp99
CTE_emp99
```

```
alpha=0.05
```

```
VaR_emp95 <- quantile(loss, 1-alpha)
CTE_emp95 <- mean(loss[loss > VaR_emp95], na.rm = TRUE)
VaR_emp95
CTE_emp95
```

```
# Define principal and participation rate
principal <- 100
p_rate <- 0.8 # 80% participation in upside
payoff <- principal * (1 + pmax(0, p_rate * nifty$returns))
```

```
# Step 2: Convert to percentage returns relative to principal
mld_returns <- (payoff - principal) / principal
```

```
# Step 3: Define losses (for risk measures, loss = -return)
losses <- -mld_returns;losses
```

```
# Step 4: Compute VaR and CTE at 95% and 99% levels
VaR_95p<- quantile(losses, 0.95);VaR_95p
VaR_99p <- quantile(losses, 0.99);VaR_99p
VaR_95p <- quantile(mld_returns, 0.05);VaR_95p
CTE_95p <- mean(losses[losses >=VaR_95p]);CTE_95p
CTE_99p <- mean(losses[losses >= VaR_99p]);CTE_99p
```

```
# Step 3: Define MLD parameters
principal <- 100
p_rate <- 0.8 # participation rate (80% of underlying movement)
```

```
# Step 4: Compute MLD payoffs and losses (no principal protection)
payoff <- principal * (1 + p_rate * returns)
losses <- (principal - payoff) / principal # positive = loss, negative = gain
```

```
# Step 5: Compute historical VaR and CTE (non-parametric)
VaR_95n<- quantile(losses, 0.95, na.rm = TRUE);VaR_95n
CTE_95n <- mean(losses[losses > VaR_95n], na.rm = TRUE);CTE_95n
```

```
VaR_99n <- quantile(losses, 0.99, na.rm = TRUE);VaR_99n
CTE_99n <- mean(losses[losses > VaR_99n], na.rm = TRUE);CTE_99n
```

Risk Analysis Using Monte-Carlo Simulation

```

set.seed(123)
n_sim <- 100000
sim_returns <- rghyp(n_sim, object = fit_ghyp)

loss_direct <- -sim_returns; loss_direct

#1.
principal <- 100
p_rate <- 0.8

payoff_pp <- principal * (1 + pmax(0, p_rate * sim_returns))
loss_pp <- (principal - payoff_pp) / principal

#3
payoff_np <- principal * (1 + p_rate * sim_returns)
loss_np <- (principal - payoff_np) / principal

alpha <- c(0.95, 0.99)

getRisk <- function(losses, alpha = c(0.95, 0.99)) {
  VaR <- quantile(losses, alpha, na.rm = TRUE)
  CTE <- sapply(VaR, function(v) mean(losses[losses >= v], na.rm = TRUE))
  data.frame(VaR_95 = VaR[1], VaR_99 = VaR[2],
             CTE_95 = CTE[1], CTE_99 = CTE[2])
}

df_MC <- rbind(
  getRisk(loss_direct),
  getRisk(loss_pp),
  getRisk(loss_np)
)

rownames(df_MC) <- c(
  "DIRECT INVESTMENT",
  "PRINCIPAL PROTECTED",
  "NON-PRINCIPAL PROTECTED"
)

df_MC

```